

AUTOMITRE • SLIDE 01

autoMITRE: AI-Powered Threat Intelligence & Mitigation System

A Unified Platform for Automated Threat Analysis and SOC Orchestration

Bridging Static Risk Design with Active, Framework-Aligned Cyber Defense

B.Sc. Graduation Project — Arab Academy for Science, Technology & Maritime Transport

Supervisor: Assoc. Prof. Dr. Ahmed Maher Moustafa



OPENING

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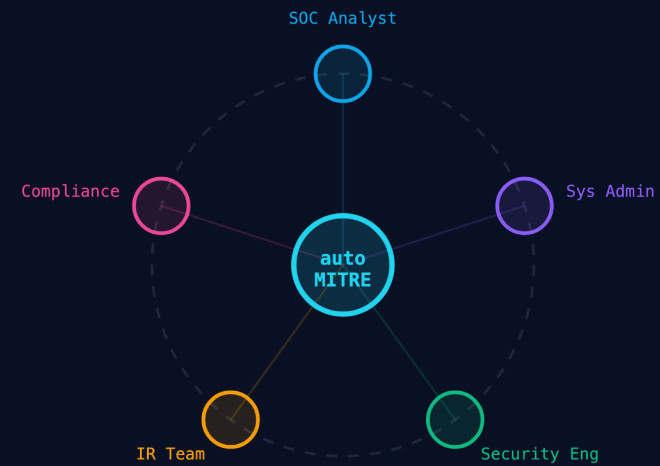
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AASTMT Smart Village | 2025–2026

WHO BENEFITS FROM AUTOMITRE



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Agenda — What We Will Cover Today

- ▶ ① The Problem — Why existing SOC workflows are broken
- ▶ ② The Gap — Where current tools fall short
- ▶ ③ Requirements & Design — How we engineered the solution
- ▶ ④ The AI Engine — SecureBERT, RAG, LLM, and the full stack
- ▶ ⑤ Results & Testing — 96.81% accuracy, 37 test cases, IEEE 829
- ▶ ⑥ Conclusion & Future Work — What comes next

PRESENTATION ROADMAP

**1. The Problem**

SOC workflow limits & alert fatigue

**2. The Gap**

Where current security tools fail

**3. Design**

Engineering & architecture choices

**4. AI Engine**

SecureBERT, RAG, & LLM flow

**5. Results**

96.8% accuracy & IEEE 829

**6. Conclusion**

Future work & DBSCAN roadmap

OPENING

The Threat Landscape is Evolving Faster Than Defenses

- ▶ 562% surge in global cyber incidents since 2019 — attacks outpace defenses.

- ▶ APTs, ransomware, and insider threats demand multi-layered, continuous analysis.

- ▶ Manual analysis cannot scale to the volume and velocity of modern threats.

GLOBAL CYBER INCIDENTS (THOUSANDS/YEAR)

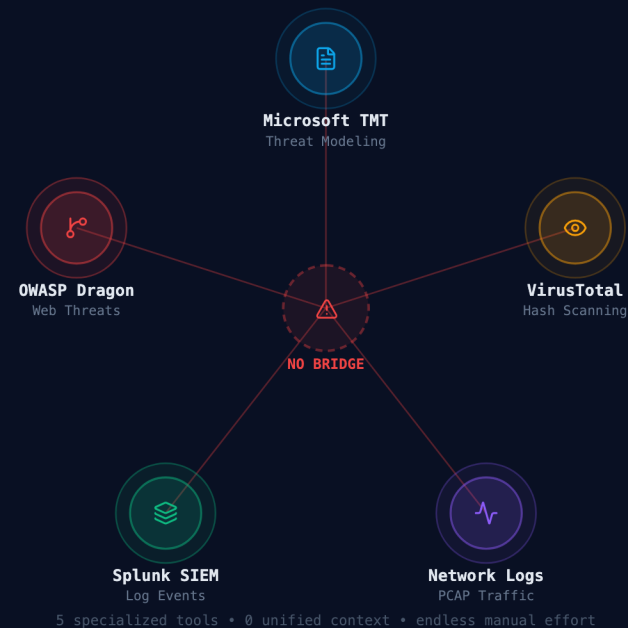
32K	48K	67K	84K	117K	158K	212K
2019	2020	2021	2022	2023	2024	2025

- +562% surge since 2019 — defenses haven't kept pace

THE PROBLEM & GAP

Fragmented SOC Tooling — The Root Cause

- ▶ Microsoft TMT, OWASP Dragon, Network Logs, VirusTotal, and Splunk all operate in isolation.
- ▶ No tool bridges design-time threat models with runtime log analysis.
- ▶ Analysts manually correlate findings across 5+ disconnected platforms every incident.



THE PROBLEM & GAP

Alert Fatigue — The SOC Analyst Bottleneck

- ▶ 10,000+ daily alerts per analyst — the human filtering ceiling has been exceeded.
- ▶ High false-positive rates cause genuine threats to be buried in noise.
- ▶ MTTD (Mean Time to Detect) averages 194 days in the industry (2024) — far too slow.

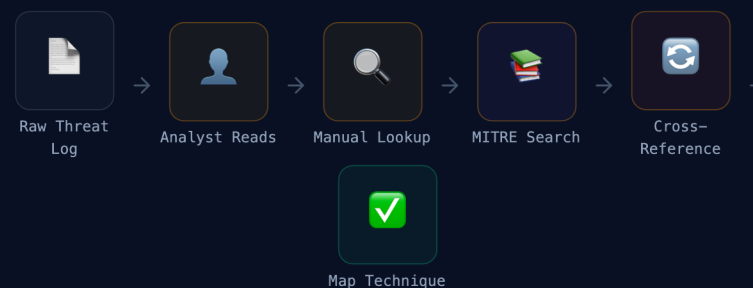
10,000+
alerts per day • 1 SOC analyst

Analyst Capacity:  OVERLOADED

THE PROBLEM & GAP

Manual MITRE Mapping — A Bottleneck in Every Incident

- ▶ Analysts spend 15–40 minutes per incident manually mapping to ATT&CK techniques.
- ▶ MITRE ATT&CK contains 650+ unique techniques and sub-techniques requiring domain expertise.
- ▶ During active incidents, this delay directly increases Mean Time to Respond (MTTR).



Avg. time per manual mapping: 15–40 minutes
Per analyst • Per incident • While attackers move in seconds

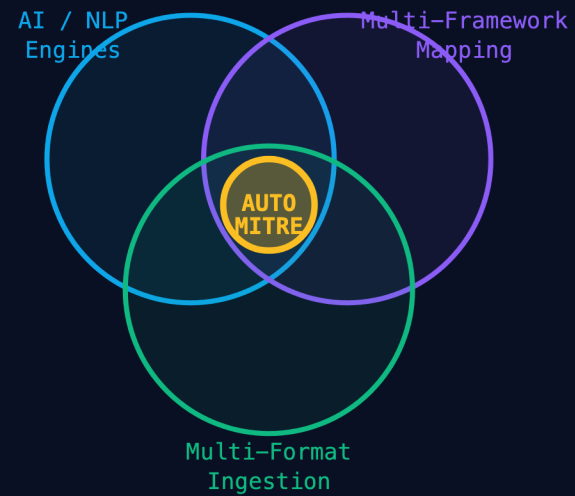
autoMITRE: same mapping in < 1.2 seconds automatically

THE PROBLEM & GAP

Problem Statement — The Exact Research Gap

- ▶ No unified platform exists that combines AI, multi-format ingestion, and multi-framework mapping.
- ▶ Current tools are reactive: they report past events but cannot predict future attack paths.
- ▶ Intelligence outputs are not SIEM-compatible — requiring additional manual reformatting.

THE RESEARCH GAP — NO TOOL COVERS ALL THREE



★ autoMITRE is the only system at the intersection of all three

LITERATURE REVIEW

Reviewing the State of the Art

- ▶ Microsoft Threat Modeling Tool: Excellent for STRIDE, but purely static and lacks runtime log analysis.

- ▶ OWASP Threat Dragon: Open-source, diagram-first, but relies heavily on manual human input.

- ▶ VirusTotal / OSINT Platforms: Strong signature detection but zero contextual mapping to MITRE T-Codes.

- ▶ Core Philosophy: autoMITRE acts as an enhancement layer to integrate with these existing tools, not a replacement for them.



REQUIREMENTS & DESIGN

autoMITRE Contribution — Filling the Gap

- ▶ Unified Intelligence: One platform ingests 7 input formats and maps to 4 frameworks.
- ▶ AI + Grounded RAG: SecureBERT embeddings anchored to verified framework definitions.
- ▶ Privacy-First: Full offline operation — no sensitive data ever sent to public clouds.

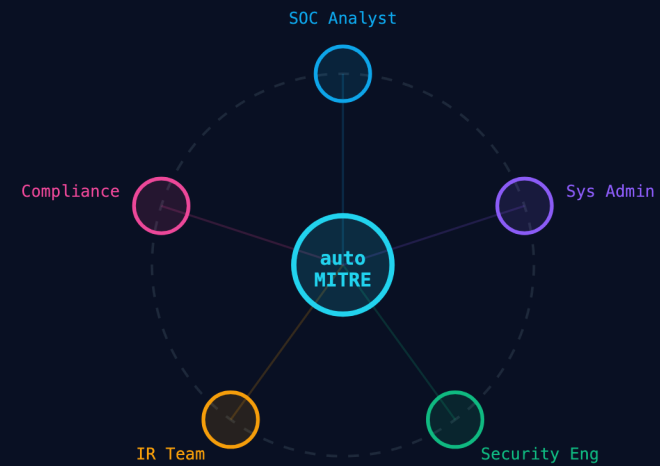


REQUIREMENTS & DESIGN

System Stakeholders — Who Benefits

- ▶ Primary: SOC Analysts and System Administrators benefit from automated threat correlation.
- ▶ Secondary: IR Teams, Security Engineers, and Compliance Officers gain explainable outputs.
- ▶ Technical: Developers and AI Engineers can extend the modular pipeline architecture.

WHO BENEFITS FROM AUTOMITRE



REQUIREMENTS & DESIGN

Functional Requirements (IEEE 830) — The 9 Core Capabilities

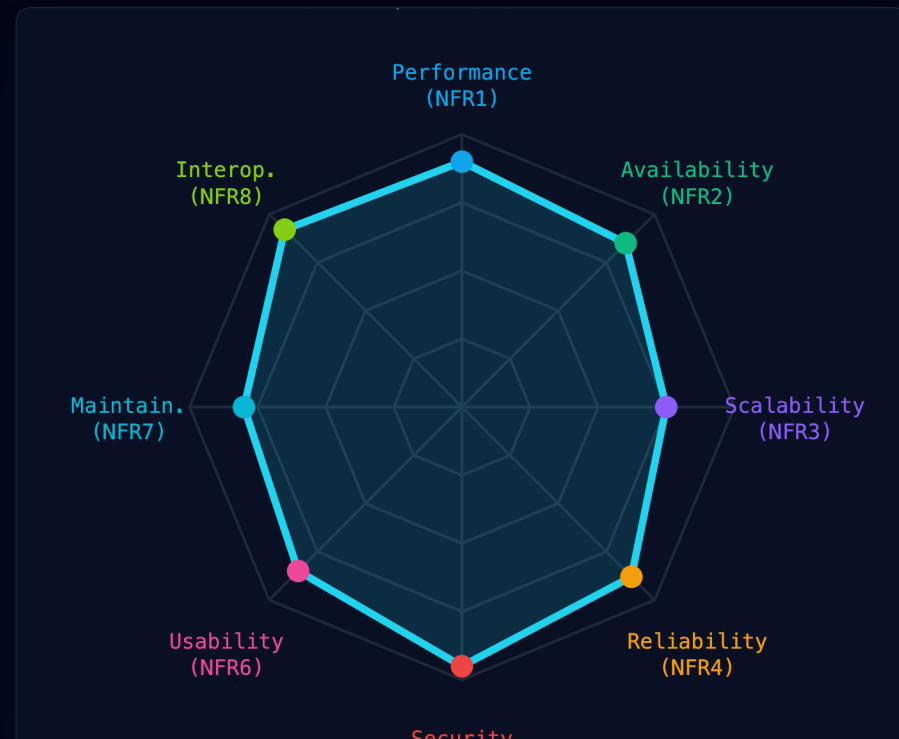
- ▶ FR1: Secure authentication & RBAC (Analyst / Administrator roles).
- ▶ FR2–FR9: Multi-format upload → AI analysis → Framework mapping → Mitigation gen → SIEM export.
- ▶ All 9 FRs were formally validated with 37 test cases — 100% PASS rate.



REQUIREMENTS & DESIGN

Non-Functional Requirements (IEEE 830) — Quality Attributes

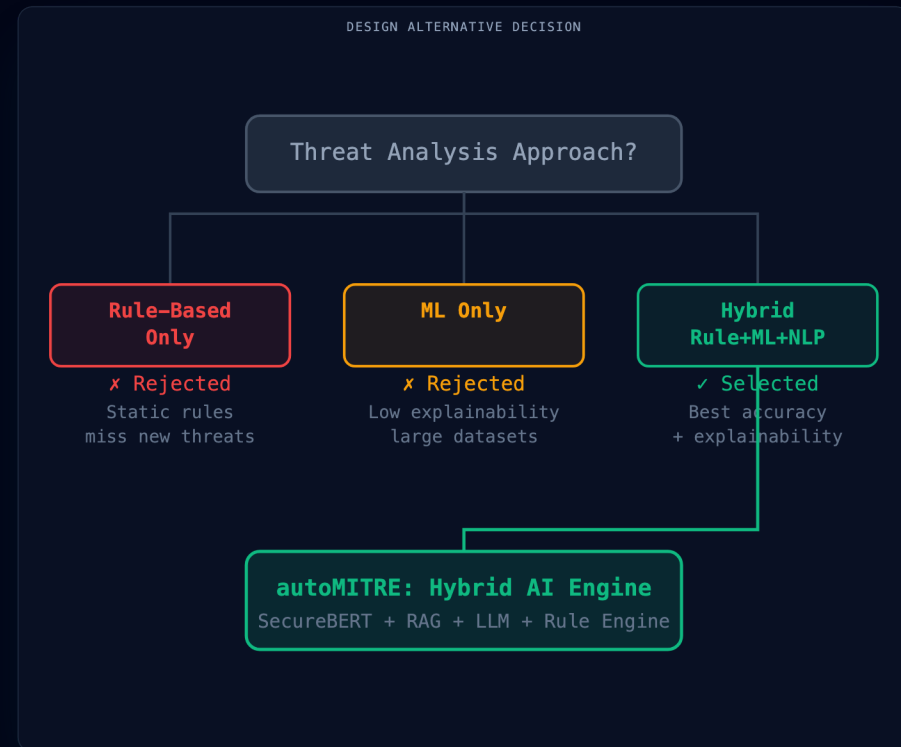
- ▶ NFR1 Performance: Cloud inference < 1.2s; local inference 4–6.5s (within hardware limits).
- ▶ NFR5 Security: Input sanitization (95% coverage), OWASP ZAP scan validation, no data leakage.
- ▶ NFR8 Interoperability: Outputs validated for Splunk, QRadar, Wazuh, and Elastic Security.



REQUIREMENTS & DESIGN

Design Alternatives Evaluated

- ▶ Alt 1 — Rule-Based Only: Rejected. High explainability but poor unknown-threat coverage.
- ▶ Alt 2 — ML Only: Rejected. Requires large labeled datasets and poor interpretability.
- ▶ Selected — Hybrid Rule + ML + NLP: Best accuracy, explainability, and academic feasibility.



REQUIREMENTS & DESIGN

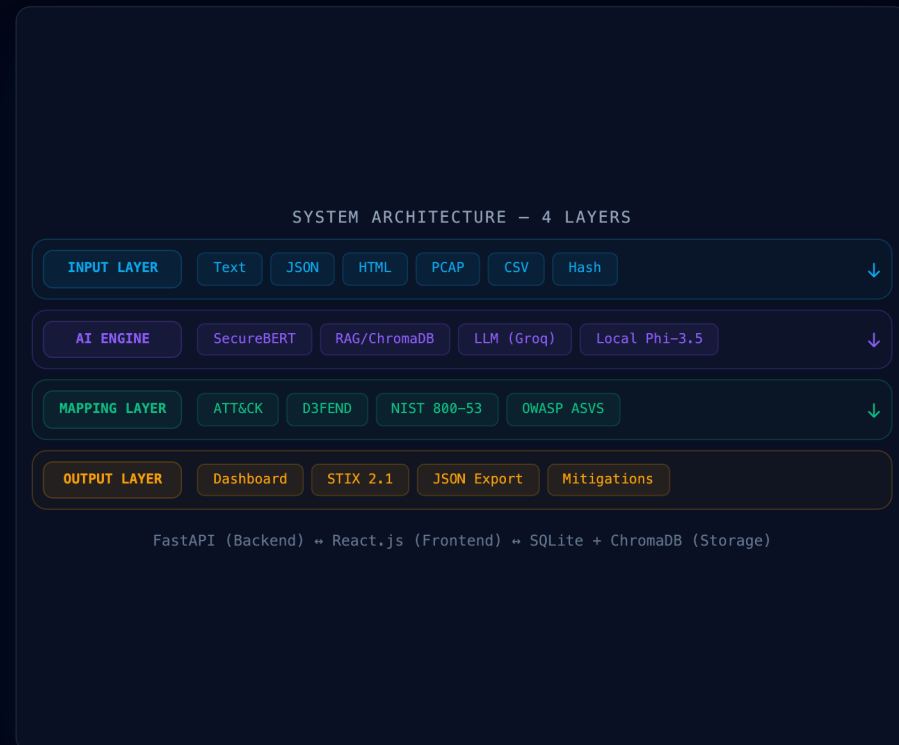
High-Level System Architecture — 4 Layers

▶ Input Layer: 7 heterogeneous formats normalized into a unified JSON threat schema.

AI Engine: SecureBERT + ChromaDB RAG (all-mpnet-base-v2) + Groq LLM + local Phi-3.5 / Llama-3.2 extractors.

▶ Mapping Layer: Cosine similarity engine correlating threats to 4 frameworks simultaneously.

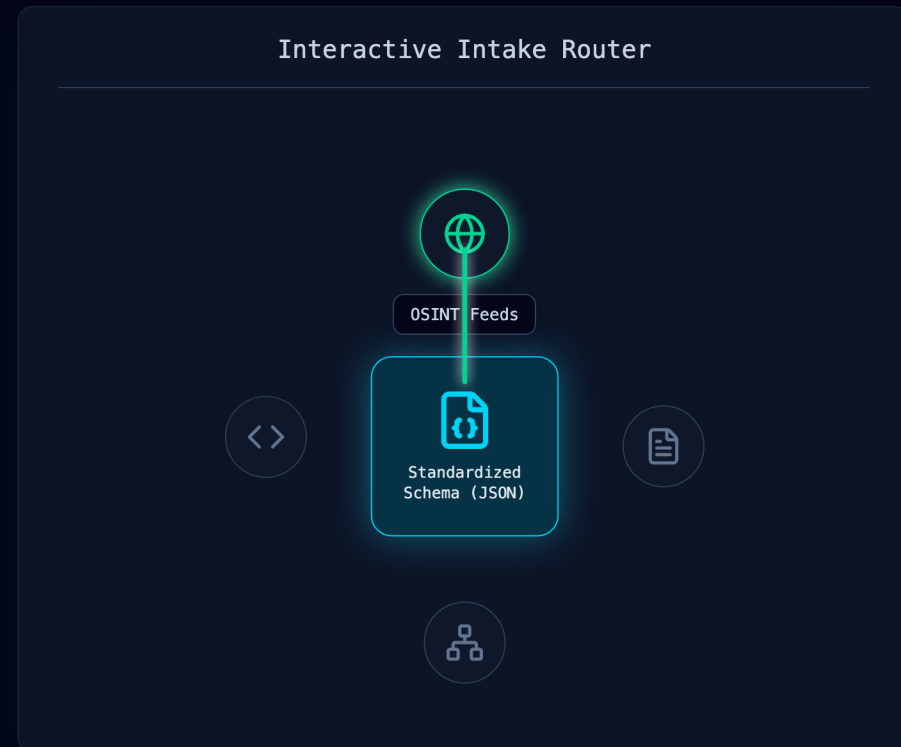
▶ Output Layer: React dashboard, STIX 2.1, JSON, and PDF exports.



THE AI ENGINE

Multi-Format Data Ingestion Layer

- ▶ Text Descriptions: Raw CTI reports, analyst notes, and threat narratives.
- ▶ Structured Inputs: IriusRisk HTML (dual-schema), JSON threat models, CSV/Excel tabular data.
- ▶ Binary Inputs: PCAP network captures (Scapy), MD5/SHA256 malware hashes (VirusTotal API).



THE AI ENGINE

Architectural Data Flow & Model Orchestration

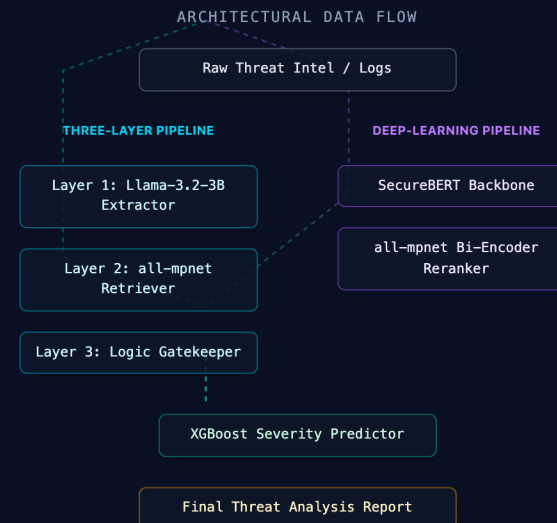
Two Parallel Paths: Unstructured text uses the 3-Layer

- ▶ RAG Pipeline; structured logs can route to Deep-Learning.

Llama-3.2 + all-mpnet: Act as the semantic extraction and retrieval engine for zero-shot framework mapping.

SecureBERT + Bi-Encoder: Provide hierarchical multi-label classification and semantic verification.

XGBoost Severity: Both paths ultimately converge into the final risk regression model.

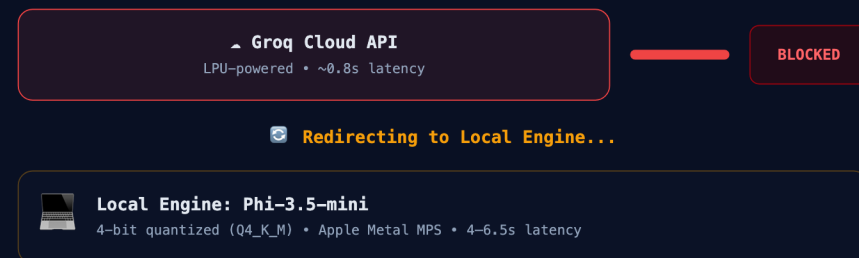


THE AI ENGINE

LLM Reasoning — Cloud Engine + Local Failover

- ▶ Primary: Groq Cloud API (LPU-powered) delivers reasoning at ~0.8s average latency.
- ▶ Fallback / Local Extraction: Phi-3.5-mini and Llama-3.2-3B-Instruct running locally via Apple Metal (MPS).
- ▶ Zero Downtime: Automatic exception-based routing — no analyst intervention required during failover.

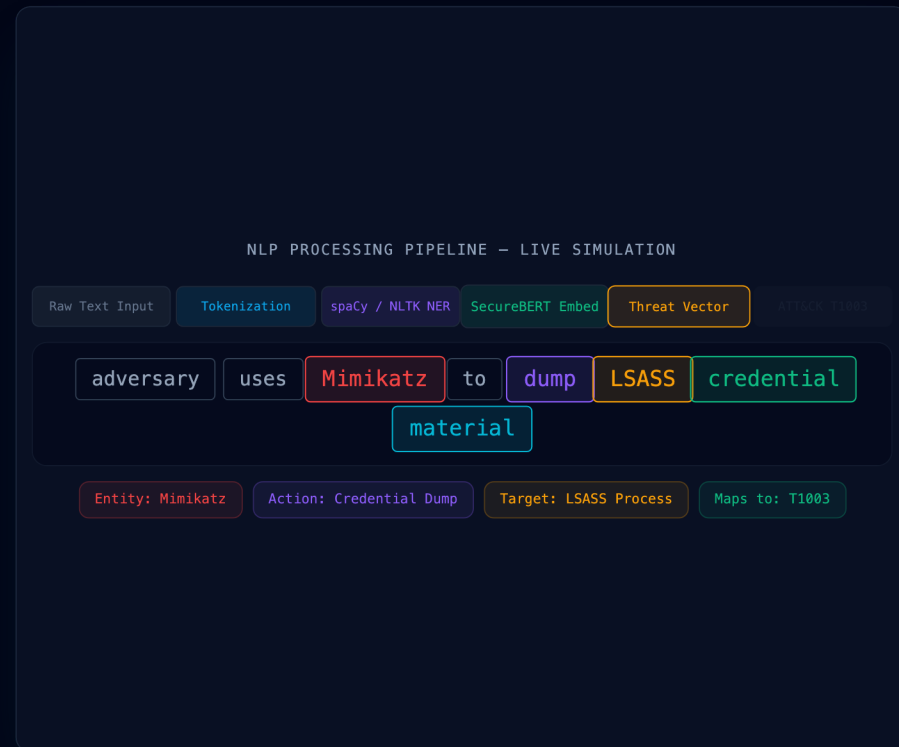
CLOUD + LOCAL FAILOVER ARCHITECTURE



THE AI ENGINE

Text & Threat Description Pipeline — NLP Processing

- ▶ Tokenization: spaCy and NLTK preprocess unstructured text into semantic tokens.
- ▶ Extraction: Llama-3.2-3B-Instruct (via MLX) acts as a local NLP relation extractor.
- ▶ Embedding: all-mpnet-base-v2 generates 768-dimensional semantic threat vectors.



THE AI ENGINE

RAG — Retrieval-Augmented Generation for Grounded Outputs

▶ Vector Database: ChromaDB stores verified ATT&CK, D3FEND, NIST, and OWASP definitions.

▶ Semantic Retrieval: Cosine similarity finds top-k most relevant framework entries per threat.

▶ Hallucination Prevention: LLM outputs are anchored to retrieved framework context — no fabricated IDs.

RAG — RETRIEVAL-AUGMENTED GENERATION

1. User Query

"Mimikatz dumps credentials from LSASS"

2. Embed → semantic vector



3. ChromaDB Search

✓ T1003 — Credential Dump

✓ LSASS Memory Access

✓ Mimikatz Tool Usage

D3FEND: Process Isolation

NIST AC-6 Controls



4. Grounded Output

Technique: T1003.001
Tactic: Credential Access
D3FEND: Process Isolation
NIST: AC-6 (Least Privilege)

THE AI ENGINE

Multi-Framework Mapping Engine

- ▶ Single-Pass Correlation: One threat maps to ATT&CK, D3FEND, NIST 800-53, and OWASP ASVS.
- ▶ Three-Layer Pipeline: Llama 3.2 Extractor → all-mpnet Retriever → Deterministic Logic Gatekeeper.
- ▶ Accuracy: 84.06% for ATT&CK (semantic) and 98.41% for D3FEND/NIST (deterministic).



THE AI ENGINE

SecureBERT — The Domain-Specific AI Classifier

Cybersecurity-Trained: Pre-trained on 20,736 threat narratives from the MITRE CTID TRAM dataset across all 499 ATT&CK techniques.

Superior Performance: 96.81% accuracy, 96.65% F1-Score — outperforms generic BERT models.

Severity Scoring: Classifies threats into CVSS-aligned severity levels at 95.22% F1-Score.

SECUREBERT — CYBERSECURITY-TUNED TRANSFORMER



THE AI ENGINE

Semantic Bi-Encoder Reranker

- ▶ Hallucination Filter: SecureBERT can occasionally output high-confidence false positives.
- ▶ Semantic Verification: all-mpnet-base-v2 cross-checks SecureBERT predictions against definitions.
- ▶ Precision Boost: Only tags with > 0.45 cosine similarity are retained, filtering out statistical noise.

HALLUCINATION FILTER PIPELINE

1. SecureBERT Output

T1059.001 (PowerShell)	Conf: 0.92
T1078 (Valid Accounts)	Conf: 0.85
T1543.003 (Windows Service)	Conf: 0.78



all-mpnet-base-v2
Threshold > 0.45

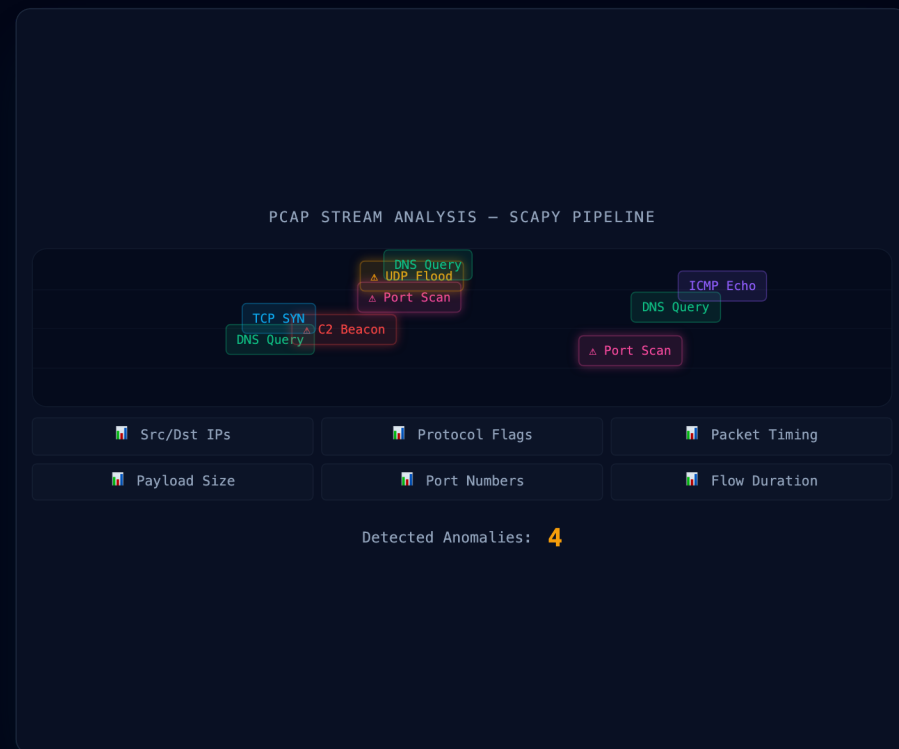
2. Semantic Verification

✓ T1059.001 (PowerShell)	Sim: 0.88
✗ T1078 (Valid Accounts)	Sim: 0.21
✓ T1543.003 (Windows Service)	Sim: 0.65

THE AI ENGINE

Isolated Pipeline — PCAP Network Analysis

- ▶ Packet Parsing: Scapy performs header inspection and reconstructs TCP/UDP flow sessions.
- ▶ Feature Extraction: src/dst IPs, ports, protocols, flags, payload size, inter-packet timing.
- ▶ Behavioral Classification: Detects beaconing, port scanning, and C2 patterns.



THE AI ENGINE

Threat Severity Prediction

► XGBoost Regressor: Predicts threat severity on a CVSS v3.1 aligned scale (0.0 - 10.0) in microseconds.

Features: 15-dimensional feature vector containing text
► complexity, entity counts, registry/process activity, and C2 indicators.

Blended Scoring: Merges XGBoost, SecureBERT, and
► heuristic scores using MAX aggregation to prevent risk dilution.

XGBOOST RISK CONVERGENCE

Both the NLP pipeline and the Deep-Learning pipeline feed into the final severity predictor to calculate a CVSS v3.1 aligned score.

RAG Pipeline

Base CVSS of Mapped Techniques

SecureBERT

Multi-label Severity Flags



XGBoost Regressor

FINAL RISK SCORE

8.4

HIGH SEVERITY

THE AI ENGINE

Predictive Attack Path Generation

► Proactive Intelligence: Transitions the system from reactive reporting to proactive forecasting.

► Kill-Chain Context: Analyzes the current tactic (e.g., Privilege Escalation) to predict logical next steps.

► LLM Reasoning: Generates explainable, probabilistic narratives of the adversary's most likely targets.

PREDICTIVE ATTACK PATH GENERATION

The LLM analyzes the current kill-chain position and generates probabilistic next-step actions the adversary is likely to take.



HISTORICAL
Initial Access



HISTORICAL
Execution



DETECTED
Privilege
Escalation



PREDICTED
Credential
Access



PREDICTED
Lateral
Movement



LLM Generated Prediction

"Based on the execution of `whoami /priv` and `mimikatz`, the adversary has achieved privilege escalation and is currently positioned to dump LSASS memory. The most probable next step is Lateral Movement via Pass-the-Hash (T1550.002) to target the Domain Controller."

THE AI ENGINE

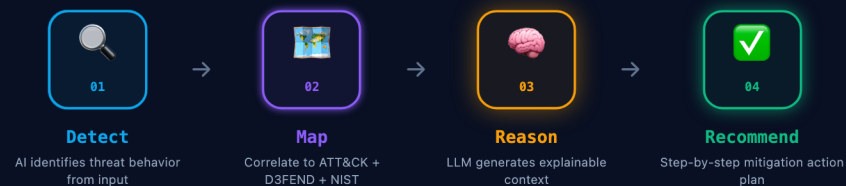
Mitigation Generation Pipeline

▶ Step 1 Detect: AI identifies threat intent, target, and attack technique from input.

▶ Step 2 Map: Correlate to ATT&CK, D3FEND, and NIST — produce defensive control IDs.

▶ Steps 3–4 Reason & Recommend: LLM generates explainable step-by-step mitigation actions.

MITIGATION GENERATION — 4-STAGE PIPELINE

**Example Mitigation Output:**

1. Enable LSASS process protection (Windows Credential Guard) [NIST AC-6]
2. Block Mimikatz signatures at endpoint [D3FEND: Software-based Process Isolation]
3. Enable Privileged Access Workstations (PAW) for domain admins [NIST SC-28]

THE AI ENGINE

FastAPI Async REST Architecture

► Asynchronous Endpoints: All routes use `async/await` — no blocking I/O during file analysis.

► Modular Routers: Isolated pipelines for text, PCAP, file hash, threat DB, and SIEM export.

► Performance Target: Met NFR1 — API responses at sub-second for most endpoints.

FASTAPI ASYNC REST ARCHITECTURE

POST /api/analyze/text → text_pipeline.py < 1.2s

POST /api/analyze/pcap → pcap_parser.py < 5s

POST /api/analyze/file → file_router.py < 2s

GET /api/threats → crud.py < 10ms

POST /api/export/stix → siem_exporter.py < 0.5s

GET /api/stats/trends → stats_router.py < 50ms

Python 3.13 • Uvicorn ASGI • Async/Await throughout • SQLAlchemy ORM

THE AI ENGINE

Technology Stack — Tools & Libraries

▶ Backend: Python 3.13, FastAPI, Uvicorn, SQLAlchemy ORM, aiosqlite, SQLite.

▶ AI/ML: SecureBERT, all-mpnet-base-v2, ChromaDB, llama_cpp, Groq API, MLX/Metal.

▶ Frontend: React.js 18, Vite, TypeScript, Recharts, Framer Motion, Lucide Icons.

TECHNOLOGY STACK

● Backend

Python 3.13 FastAPI Uvicorn
SQLAlchemy SQLite aiosqlite

● AI / ML

SecureBERT all-mpnet-base-v2
ChromaDB (RAG) llama_cpp Groq API
MLX / Metal

● NLP

spaCy NLTK Hugging Face
scikit-learn Scapy stix2

● Frontend

React.js 18 Vite TypeScript
Recharts Framer Motion Lucide

RESULTS & TESTING

Testing Framework — IEEE 829 Standard

- ▶ Test Plan Structure: Three domains — Unit/Integration, AI Quantitative, and E2E System testing.
- ▶ Test Coverage: 37 formal test cases (TC-01 to TC-37) spanning all 9 Functional Requirements.
- ▶ Outcome: 37 / 37 PASS (100%) — 0 Failures, 0 Blocked, 0 Deferred.

IEEE 829 TEST CASE MATRIX (37 TOTAL)

TC-01 ✓	TC-02 ✓	TC-03 ✓	TC-04 ✓	TC-05 ✓	TC-06 ✓
TC-07 ✓	TC-08 ✓	TC-09 ✓	TC-10 ✓	TC-11 ✓	TC-12 ✓
TC-13 ✓	TC-14 ✓	TC-15 ✓	TC-16 ✓	TC-17 ✓	TC-18 ✓
TC-19 ✓	TC-20 ✓	TC-21 ✓	TC-22 ✓	TC-23 ✓	TC-24 ✓
TC-25 ✓	TC-26 ✓	TC-27 ✓	TC-28 ✓	TC-29 ✓	TC-30 ✓
TC-31 ✓	TC-32 ✓	TC-33 ✓	TC-34 ✓	TC-35 ✓	TC-36 ✓
TC-37 ✓					

● 37 / 37 PASS (100%) | 0 Failures • 0 Blocked

RESULTS & TESTING

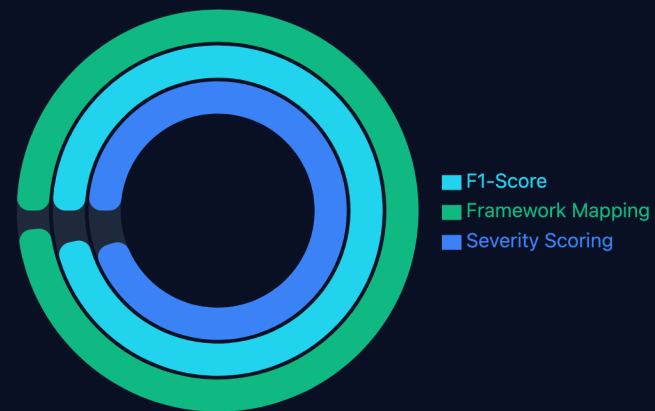
AI Model Accuracy — SecureBERT Evaluation

Dataset: MITRE CTID TRAM dataset (20,736 cyber threat narratives across all 499 MITRE ATT&CK techniques).

Core Classification: 96.81% Accuracy and 96.65% F1-Score.

Severity Scoring: 95.22% F1-Score for CVSS-aligned threat severity classification.

SecureBERT Accuracy Metrics (%)

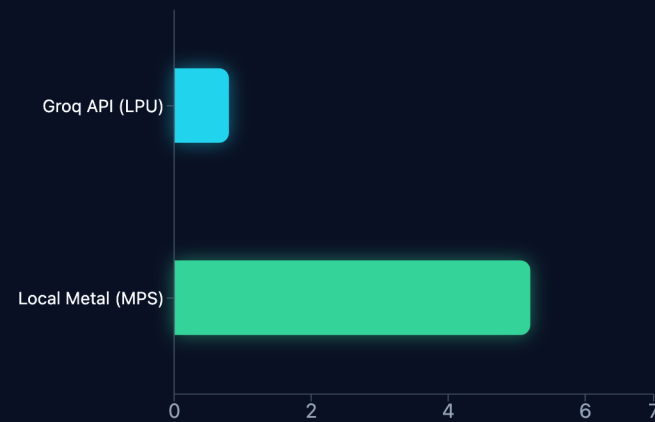


RESULTS & TESTING

Inference Latency Benchmarks

- ▶ Groq Cloud (Primary): ~0.8–1.2 seconds average per request using LPU-powered inference.
- ▶ Local Phi-3.5 (Fallback): 4.0–6.5 seconds depending on prompt length via Apple Metal MPS.
- ▶ Both modes meet NFR1 performance targets — fully operational within hardware constraints.

Inference Latency (Seconds)



RESULTS & TESTING

Framework Mapping Accuracy

- ▶ ATT&CK Semantic Mapping: 84.06% accuracy — semantic cosine similarity across 499 techniques.

- ▶ D3FEND + NIST 800-53: 98.41% accuracy — deterministic rule-based framework correlation.

- ▶ A single detected threat reliably populates all four frameworks in one automated pass.

FRAMEWORK MAPPING ACCURACY

**ATT&CK Technique Mapping**

499 techniques tested

**D3FEND + NIST Mapping**

Deterministic mapping

ATT&CK semantic matching uses cosine similarity (harder problem – 499 unique techniques)
D3FEND/NIST uses deterministic framework rule mapping (near-perfect coverage)

RESULTS & TESTING

Contextualizing 84% Accuracy

Sub-technique Ambiguity: 499 techniques have

- ▶ massive semantic overlap (e.g., T1003.001 vs T1003.004).

Probabilistic Nature: Zero-shot semantic search

- ▶ retrieves based on generic embeddings, lacking domain nuance.

Isolated Context: Logs are analyzed in a vacuum.

- ▶ Analysts use surrounding timeline context to map correctly.

THE AMBIGUITY PROBLEM

INPUT LOG (MISSING CONTEXT)

"Suspicious process accessed credential storage"

↓ Semantic Search ↓

T1003.001

LSASS Memory

Similarity: 82%

T1003.002

SAM Database

Similarity: 79%

T1003.004

LSA Secrets

Similarity: 81%

The AI correctly identifies the T1003 Tactic, but struggles to pick the exact sub-technique without surrounding timeline context.

RESULTS & TESTING

Case Study — Mimikatz Credential Dumping

► Input: "adversary uses Mimikatz to dump credential material from LSASS"

► Mapped to: T1003.001 (OS Credential Dumping) | D3FEND: Process Isolation | NIST AC-6

► Mitigation: Credential Guard + Mimikatz signature blocking + PAW — generated in 1.2 seconds.

LIVE CASE STUDY: MIMIKATZ CREDENTIAL DUMPING

01 **Raw Input**
"adversary uses Mimikatz to dump credential material from LSASS" ✓

02 **SecureBERT Extraction**
Entity: Mimikatz | Action: credential dump | Target: LSASS process ✓

03 **ATT&CK Framework Mapping**
T1003 – OS Credential Dumping | Tactic: Credential Access ✓

04 **Defensive Framework Mapping**
D3FEND: Credential Access Prevention | NIST AC-6: Least Privilege ✓

05 **Mitigation Output**
1. Enable Credential Guard 2. Block Mimikatz signatures 3. Implement PAW

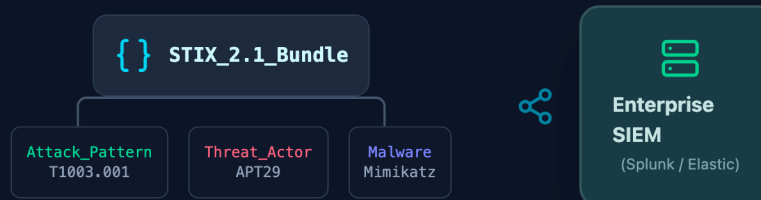
Total end-to-end time: ~1.2 seconds (Groq cloud mode)

RESULTS & TESTING

SIEM Integration — STIX 2.1 Export Validation

- ▶ STIX 2.1 Objects: Detected threats are structured as Bundle → Indicator → AttackPattern objects.
- ▶ Compatibility: Validated ingestion into Splunk Enterprise Security and Wazuh dashboards.
- ▶ Export Formats: STIX 2.1 JSON, plain JSON, and CSV — covering all major SIEM platforms.

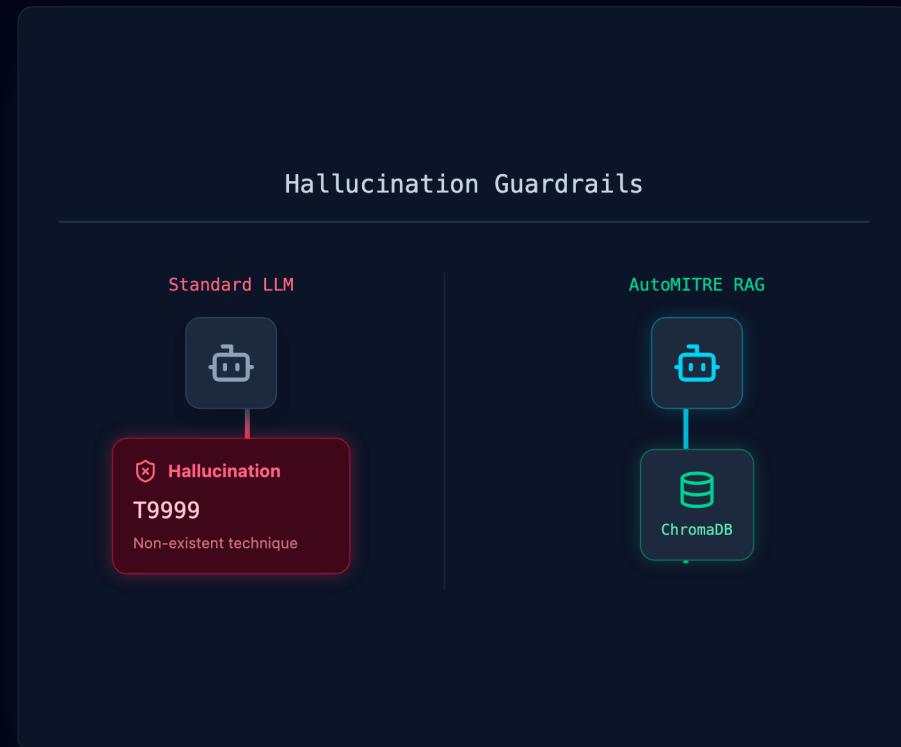
STIX 2.1 Export & Splunk Interoperability



RESULTS & TESTING

Hallucination Prevention — RAG Validation

- ▶ Problem: Generic LLMs hallucinate non-existent ATT&CK IDs and fabricate defense mappings.
- ▶ RAG Solution: Every LLM output is anchored to retrieved ChromaDB framework definitions.
- ▶ Result: Zero hallucinated technique IDs in all 37 test runs — 100% framework-verified outputs.



RESULTS & TESTING

System Resiliency — Failover Test (TC-33)

▶ Test: Groq Cloud API artificially blocked during active threat processing session.

▶ Result: Exception handler automatically redirected to local Phi-3.5-mini within 0.3 seconds.

▶ Zero Downtime: All 12 queued threat analyses completed — no crashes, no data loss.

RESILIENCY TEST TC-33 — FAILOVER VALIDATION

- **TC-33 Started**
Normal operation — Groq API serving requests
- **API Artificially Blocked**
Network exception injected at 45s mark
- **Exception Caught**
ConnectionError detected in ai_threat_analyzer.py
- **Failover Triggered**
Redirecting to local Phi-3.5-mini via llama_cpp
- **Local Engine Active**
Metal MPS acceleration engaged — 4.2s latency
- **Threat Queue Cleared**
All 12 pending analyses completed — Zero crashes

✓ **Result: PASS — Zero downtime • Zero memory crashes**

RESULTS & TESTING

Load Testing — 20 Concurrent Analysts (Locust)

- ▶ Load Profile: Simulated 1 → 20 concurrent analysts uploading files simultaneously.
- ▶ Result: Linear latency degradation (0.8s → 5.1s) — no server timeouts or connection drops.
- ▶ Degradation Pattern: Predictable, manageable scaling within SOC analyst workstation limits.



RESULTS & TESTING

Functional Requirements Validation — Complete Coverage

► 9 Functional Requirements — all formally validated against dedicated test case ranges.

► All test cases PASS: Authentication, Upload, AI Analysis, Mapping, Mitigation, SIEM Export.

► NFRs met: Performance (< 1.2s), Security (95% sanitization & OWASP ZAP scan), Interoperability (STIX).

FUNCTIONAL REQUIREMENTS VALIDATION

FR1	Auth & RBAC	TC-01 to TC-03	✓ PASS
FR2	Multi-Format Upload	TC-04 to TC-07	✓ PASS
FR3	AI Threat Analysis	TC-08 to TC-11	✓ PASS
FR4	Framework Mapping	TC-12 to TC-13	✓ PASS
FR5	Mitigation Gen	TC-14	✓ PASS
FR6	Threat Prediction	TC-15	✓ PASS
FR7	Dashboard UI	TC-16	✓ PASS
FR8	PDF Export	TC-17	✓ PASS
FR9	SIEM / STIX Export	TC-18	✓ PASS

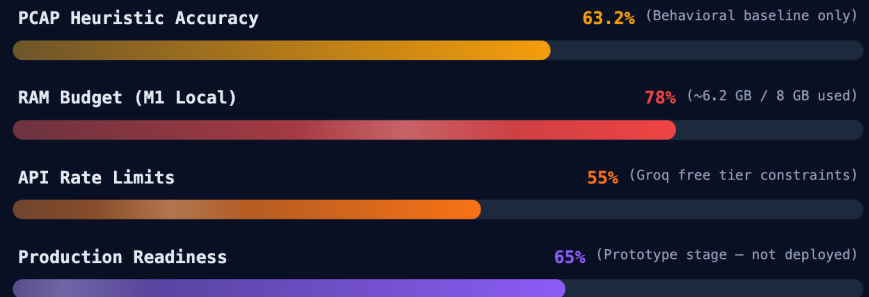
9 / 9 Functional Requirements Validated ✓

CONCLUSION & FUTURE WORK

Honest Limitations

- ▶ PCAP Heuristic Accuracy: 63.2% — behavioral baseline detection only (no deep-learning DPI yet).
- ▶ Hardware Constraint: 8GB RAM limits local model size and real-time PCAP streaming at scale.
- ▶ Production Scope: Functional prototype — not yet hardened for enterprise production deployment.

HONEST LIMITATIONS ASSESSMENT

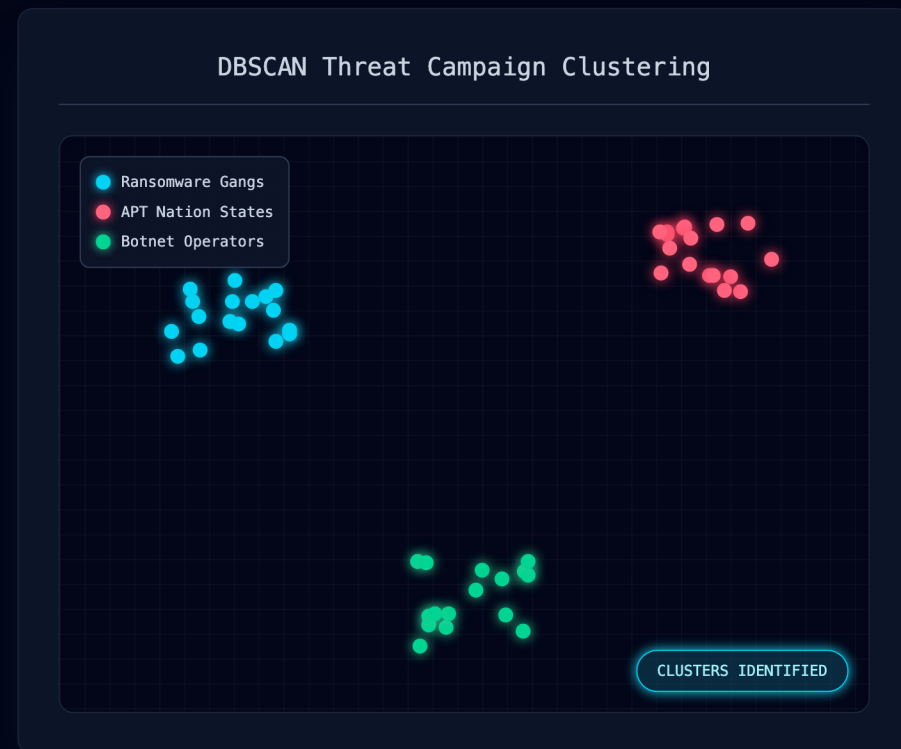


Limitations are documented honestly per academic project scope
Future work addresses each constraint

CONCLUSION & FUTURE WORK

Future Work — DBSCAN Zero-Day Anomaly Detection

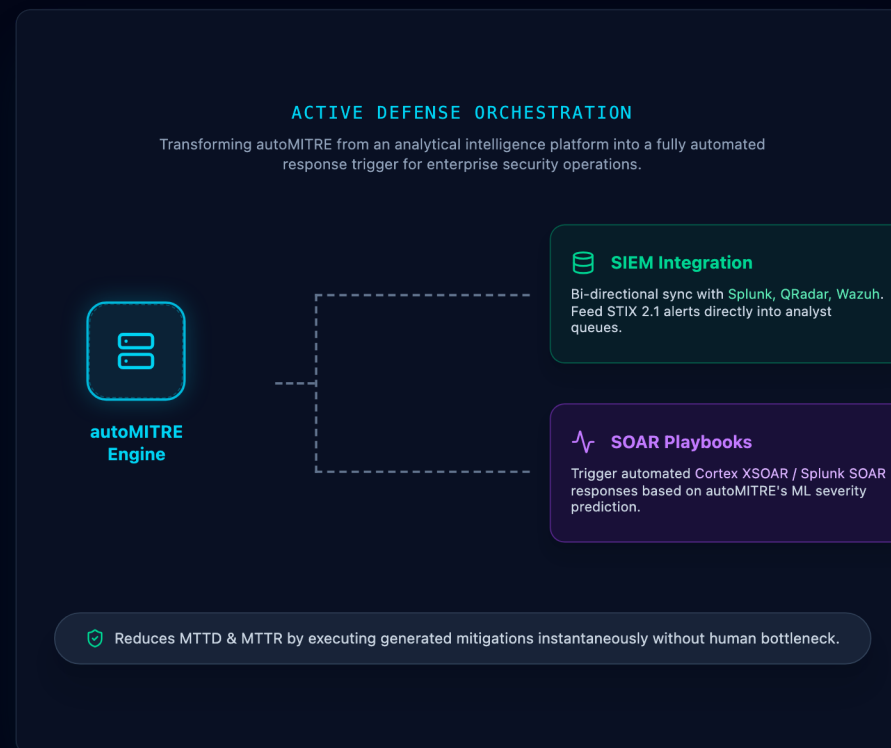
- ▶ DBSCAN Clustering: Density-based unsupervised algorithm groups behavioral anomalies.
- ▶ Zero-Day Discovery: Clusters logs that match no existing ATT&CK signature — novel threat detection.
- ▶ Self-Improving: Newly confirmed clusters feed back into the RAG knowledge base automatically.



CONCLUSION & FUTURE WORK

Future Work — SIEM & SOAR Integration

- ▶ Active Defense: Moving autoMITRE from passive intelligence to an active automated response trigger.
- ▶ SIEM Sync: Native, bi-directional API hooks for Splunk Enterprise Security and QRadar.
- ▶ SOAR Playbooks: Trigger Cortex XSOAR isolation playbooks immediately upon high-severity predictions.



CONCLUSION & FUTURE WORK

Future Enhancement Roadmap

► Phase 1 (Q3 2026): Fine-tune local models to match cloud accuracy — eliminate offline performance gap.

► Phase 2 (Q4 2026): Add Cloud Security Alliance (CSA), ISO 27001, GDPR framework mappings.

► Phase 3–4 (2027): Enterprise SecLM/SecPaLM Integration to achieve 95%+ mapping accuracy.



Conclusion & Thank You

autoMITRE: A complete AI-powered threat intelligence system — built from the ground up by a team of 6.

96.81% accuracy • 37/37 test cases PASS • 4 frameworks • SIEM-ready — all within 8GB RAM.

The floor is open for Questions & Answers. Thank you for your time and attention.

